

Developing on the Bleeding Edge *of MCP Servers*

Announcements

Introducing the Model Context Protocol

25 Nov 2024 • 3 min read

r/ClaudeAI • 1 yr. ago
JoshSummers

Can someone explain MCP to me? How are you using it? And what is the difference between you to do that you couldn't do before?

ELI5 r/explainlikeimfive • 6 days ago
spidey_ken

ELI5 . What is an MCP ? And what is the difference

Technology

How they have transformed people's Clau

r/mcp
Jaydg

r/AI_Agents • 1 mo. ago
barmic12

Can so Is anyone actually

What's an
the hype

Discussion

There have probably been ten
demos based on Claude or ot
streamlining work with some

r/4xe1

What's the point of it?

I'm a bit confused about the purpose of MCP

- Simplified Integration
- Interoperability
- Enhanced Capabilities

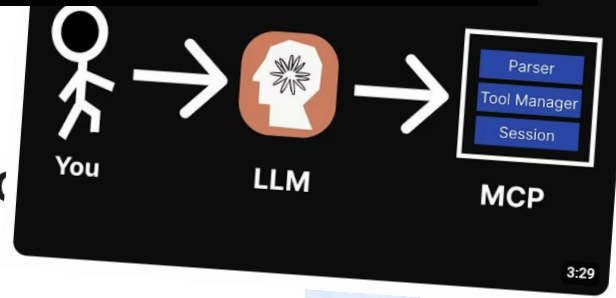
r/devsecops • 5 mo. ago
NegativePackage7819

what is an MCP and why should I

Shweta  @shweta_ai • Mar 6



- > be me
- > see post about MCP
- > don't understand it
- > bookmark it



useful tool that perhaps your LLM can make

... itself does not call the tool but rather
... return the result

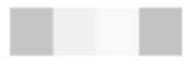
standard terms like xml

are they targeting, and

Why is MCP so hard to understand?
3.1K views • 5 months ago

Caleb Writes Code

Let's talk about why MCP is so difficult to understand.



24 Mar at 2:15 PM

MCP is popping off in all my feeds these last few weeks



MCP dev is incredibly frustrating 😅

We could do our own chat experience

Chat UI is much more wrangle-able

not the worst thing

if the user has to ask the agent to run a tool?



Claude



Claude

🔍 Active Weather Alerts for New York

🔒 Share



note the extended Winter Storm Warnings that stretch into December 13th.

Would you like me to focus on any particular region or type of alert in more detail?

ZL whats the weather in Sacramento

Let me check the current weather in Sacramento, California.

[View Result from get-weather from weather \(local\) >](#)

Currently in Sacramento:

- Temperature: 56°F (falling to around 53°F in the afternoon)
- Conditions: Haze clearing to mostly sunny
- Wind: Light northeast wind around 2 mph
- Notable conditions: Areas of frost and fog before 11am

Next few days:

- Tonight: Low of 37°F, hazy conditions
- Wednesday: High of 56°F, partly sunny with morning frost
- Wednesday Night: 50% chance of rain showers, low around 44°F
- Thursday: 50% chance of morning showers, partly sunny, high near 55°F

Would you like more details about the forecast for any particular day?

Debug Copy Retry Share



Claude can make mistakes. Please double-check responses.

ZL



Reply to Claude...



Claude 3.5 Sonnet Choose style

2

2

- Ambiguous
- Generic
- Outcome unclear

~~workflow_create~~

~~workflow_run~~

~~step_create~~

~~step_run~~

- Semantically clear
- Contextual
- Intuitive

create_spec

edit_spec

generate_code

generate_tests

run_tests

- AGENTS.md
- CLAUDE.md
- Subagents

AGENTS.md

A simple, open format for guiding coding agents, used by over 20k open-source projects.

Think of AGENTS.md as a **README for agents**: a dedicated, predictable place to provide the context and instructions to help AI coding agents work on your project.

[Explore Examples](#)[View on GitHub](#)

```
---
name: your-sub-agent-name
description: Description of when this subagent should be invoked
tools: tool1, tool2, tool3 # Optional - inherits all tools if omitted
model: sonnet # Optional - specify model alias or 'inherit'
---
```

Your subagent's system prompt goes here. This can be multiple paragraphs and should clearly define the subagent's role, capabilities, and approach to solving problems.

Include specific instructions, best practices, and any constraints the subagent should follow.

- Our MCP server kept crashing, but...
 - Breakage just looked like non-adherence
- Important: the agent can rarely introspect, or diagnose itself

What about *optimising*?

Progress

 Copy page



Protocol Revision: 2025-06-18

The Model Context Protocol (MCP) supports optional progress tracking for long-running operations through notification messages. Either side can send progress notifications to provide updates about operation status.

How to report progress with an MCP Server #4157

[Open](#)



macebake on Aug 1 · edited by macebake

Edits ▾ ...

I've also had this issue, commenting to stay informed.

I thought an alternative might be to use [progress monitoring](#), but this didn't work. Concluded that Claude Code doesn't support progress tokens -- certainly there's nothing in the docs that suggests they do yet.



coygeek on Aug 16



Hi there! Thanks for the excellent question and for providing a clear example of what you're trying to accomplish. It's a great use case.

You've correctly identified the `notifications/progress` method in the Model Context Protocol (MCP) specification. However, there's a key distinction in how the protocol works that's causing the behavior you're seeing.



coygeek on Aug 18

...

Hi [@jakepgoldman](#), [that's a fantastic question](#) and a [perfect example](#) of a real-world, complex agentic workflow. The "human-in-the-loop" pattern is a powerful but tricky one to implement. You're right on track thinking that your current polling method isn't optimal.

[Let's break this down.](#)

Directly Answering Your Timeout Question

Is there a max timeout for streaming?

Yes, absolutely. While there isn't a single "MCP max timeout" documented, a 30+ minute open streaming connection is not a viable or reliable approach. The connection would almost certainly be terminated long before that by:

- Client-side timeouts in Claude Code itself.
- Intermediate proxies or load balancers (either on your network or ours) that clean up long-lived connections.
- Simple network instability over a 30-minute period.

So, your instinct is correct: a simple, long-held streaming connection is not the solution for this kind of asynchronous, human-driven task. [Your current approach](#) of returning an ID is architecturally much closer to the right pattern.

The Core Challenge: True Asynchronicity

The problem, as you've noted, is the user experience. Making the user manually ask for an update is clunky. We need a way for the agent to gracefully handle a long pause and resume contextually.



:coygeek:

- Just because it's in the official spec...
- Just because it's supported by the MCP framework you use...
- Doesn't mean clients will support it
 - Absence of evidence is evidence of absence
- Few tools you can use to track
 - agentcompat.com

Feature	Claude Code	Cursor	VSCode (Copilot)	Codex
MCP Spec				
Transport Specifies how messages are transmitted (stdio, HTTP streaming, custom transports).	✓	✓	✓	✓
Authorization Provides OAuth-based authentication/authorization for HTTP transports.	✓	✓	✓	✓
Utilities Cross-cutting support for error handling, structured messages, and other shared mechanics.	✓	✓	✓	✓
Resources Servers expose read-only contextual data (files, database records, APIs) that clients/models can fetch.	✓	✗ *	✓	✓
Prompts Servers define reusable prompt templates with parameters that clients/models can list and invoke.	✓	✗ *	🔘	✓
Tools Servers expose structured operations or actions (functions) that clients/models can call.	✓	✓	✓	✓
Logging Servers can emit structured log messages back to the client for visibility and debugging.	✓	✓	✓	✓

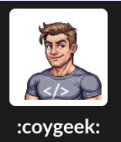
Also, don't ask the agent!



Claude Code v2.0.19
Sonnet 4.5 · API Usage Billing
/Users/macey/git/backend

> do you support mcp progress tracking?

- Yes, I do support MCP progress tracking!





What *next*?

- “Botfooding”
 - How well can an agent *use* your MCP server?
 - Ask for continuous feedback, and reflection

“Specs mostly covered the happy path and ‘not found’ errors. What about bad state, concurrency, or internal failures?”

“Explicit dependency declarations made the system easier to reason about. Circular imports were never a problem. But, I had to manually trace how everything connected. A graph would’ve helped.”

↑↓ Created At ▾

Q Search by name, description, or keywords...

⌵ K

All Tasks

127

Filesystem

30

Notion

28

Playwright

25

Github

23

Postgres

21

Transactional Inventory Tra... ★

L3

Postgres Lego

Create PostgreSQL function to handle inventory part transfers between LEGO sets with validation and audit logging.

Transactional Operations, Stored Procedures And Functions, Audit And Compliance

Created by Jiawei Wang

2025-08-16

Linting Ci Workflow ★

L3

Github MCPMark CI/CD

Set up ESLint workflow for code quality enforcement on all pull requests with proper CI integration.

CI Cd Automation, Pr Workflows

Created by Zijian Wu

2025-08-15

Contact Information ★

L3

Filesystem Desktop Template

Extract contact details from various file formats on desktop and perform reasoning analysis on the collected relationship data.

Data Extraction, Cross Referencing

Created by Lingjun Chen

2025-08-14

Change Color ★

L3

Notion Toronto Guide

Navigate to the Toronto Guide page and change all pink-colored elements to different colors.

Visual Formatting, Conditional Filtering

Created by Xiangyan Liu

2025-08-14

NBA Statistics Analysis ★

L3

Playwright Reddit

Create sports analytics account, collect NBA player statistics from forum discussions, analyze basketball performance metrics, and compile...

User Interaction, Data Extraction, Comparative Analysis, Content Submission

Created by Fanqing Meng

2025-08-12

Cloudflare Turnstile Challenge ★

L3

Playwright Eval Web

Navigate websites with Cloudflare Turnstile protection, handle security challenges, bypass bot detection mechanisms, and successfully...

User Interaction

Created by Allison Zhan

2025-07-27

Film Inventory Management

L3

Customer Analysis Fix

L3

Customer Analytics Optimizati...

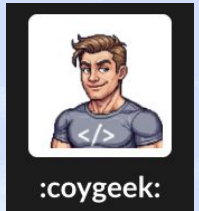
L3

Example Output:

```
<answer>
SpriteProducts|XX
Quantity100Products|XX
WS12Info|Product Name Here:$XX.XX
PendingOrders|X
GraceOrderID|00000XXXX
HighestOrderInfo|Customer Name:$XXX.XX
CheapProduct|Product Name:XX
OvernightDufflePrice|$XX.XX
HollisterPosition|Xth
CostelloCustomers|X
SarahMillerInfo|Group Name:MMM DD, YYYY
PaidInvoices|X
Invoice002BillTo|Customer Name
</answer>
```


Takeaways

- Agents are users too!
 - Design with empathy for them
 - Collect feedback from them
- Keep MCP spec & client support disparities in mind
 - Hack around them!
- Don't ask an agent to debug itself



Resources

Tool discovery musings: <https://ainativdev.io/news/teaching-mcp-servers-new-tricks>

Block's guide to MCP design: <https://engineering.block.xyz/blog/blocks-playbook-for-designing-mcp-servers>

Spec vs client support tracking:

- agentcompat.com
- <https://modelcontextprotocol.io/clients>
- <https://github.com/apify/mcp-client-capabilities>

Botfooding: <https://ainativdev.io/news/botfooding-can-an-llm-give-good-user-feedback>

MCPMark: <https://mcpmark.ai/tasks/>